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Web&Cloud
Labs10

10.1g

How long did the job take to execute?
About a minute and a half

Examine output.txt and show the estimate of π calculated.
3.1419113514191137

```
97, op_mkdirs_min=297, stream_write_b
[CONTEXT ratelimit_period="5 MINUTES"
Pi is roughly 3.1419113514191137
@@@
```

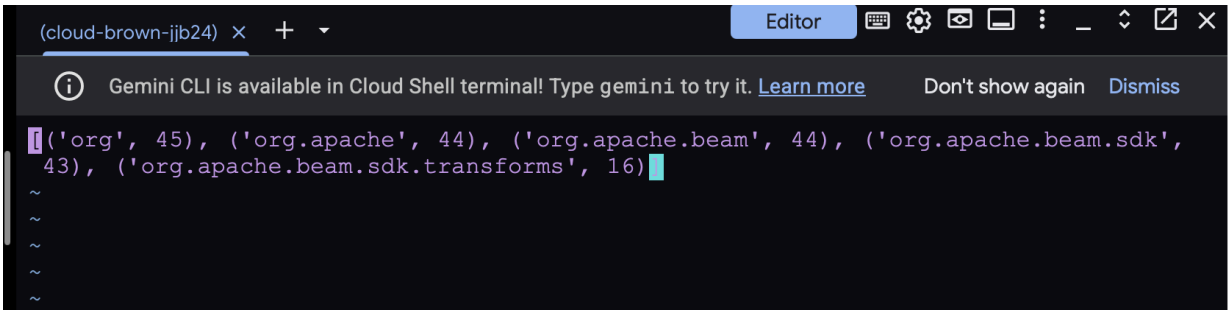
How long did the job take to execute? How much faster did it take?
About 30 seconds. A minute faster.

Examine output2.txt and show the estimate of π calculated.

```
Pi is roughly 3.1418585914185857
25/12/02 04:40:27 INFO DataprocSpark
tion_http_patch_request=0, files_cre
ist_status_result_size=0, op_open=0,
```

10.2g

- Where is the input taken from by default?
  - ../javahelp/src/main/java/com/google/cloud/training/dataanalyst/javahelp/
- Where does the output go by default?
  - /tmp/output/javahelp/src/main/java/com/google/cloud/training/dataanalyst/javahelp/

- Examine both the getPackages() function and the splitPackageName() function. What operation does the 'PackageUse()' transform implement?
    - Its a generator that processes a line of code to identify and count the uses of packages.
- Look up Beam's CombinePerKey. What operation does the TotalUseoperation implement?
    - TotalUse implements a summation of package counts.
- Which operations correspond to a "Map"?
    - 'GetImports' >> beam.FlatMap
    - 'PackageUSe' >> beam.FlatMap
- Which operation corresponds to a "Shuffle-Reduce"?
    - 'TotalUse' >> beam.CombinePerKey
- Which operation corresponds to a "Reduce"?
    - sum function that is passed to beam.CombinePerKey in the 'TotalUse' step.
- Take a screenshot of its contents
    - 

- Explain what the data in this output file corresponds to based on your understanding of the program.
    - ('org.apache', 44): This means that across all the input files, packages starting with org.apache were imported 44 times.
    - ('org.apache.beam.sdk.transforms', 16): This shows that packages within the org.apache.beam.sdk.transforms namespace were imported 16 times.

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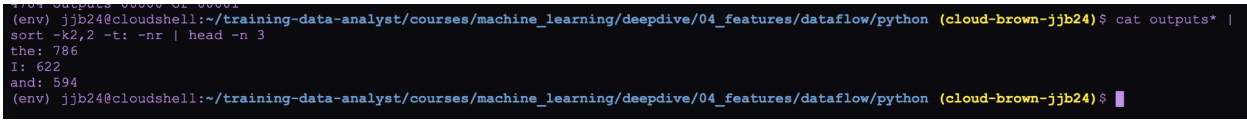
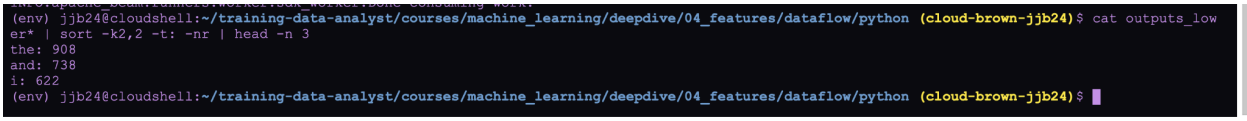
- What are the names of the stages in the pipeline?
    - Read
    - Split
    - PairWithOne
    - GroupAndSum
    - Format
    - Write
- Describe what each stage does.
    - Read: This initial stage reads the text from the input file specified by the --input command-line argument. The result is a PCollection where each element is a line from the text file.

- Split: This stage applies a parallel transformation (ParDo) that takes each line from the previous stage and splits it into individual words.
- PairWithOne: Each word is then passed to this Map transformation, which converts each word into a key-value pair. The word itself is the key, and the number one is the value (e.g., ('word', 1)).
- GroupAndSum: This stage, a CombinePerKey transformation, groups all the pairs by their unique key (the word) and then sums their corresponding values. The result is a collection of key-value pairs where the key is the word and the value is its total count.
- Format: This MapTuple stage takes the final word counts and formats them into printable strings, where each string is in the format "word: count".
- Write: The final stage, WriteToText, takes the formatted strings and writes each one as a line to the output file specified by the --output argument.

- Use wc with an appropriate flag to determine the number of different words in King Lear.
  -

- Use sort with appropriate flags to perform a *numeric* sort on the *key field* containing the count for each word in *descending* order. Pipe the output into head to show the top 3 words in King Lear and the number of times they appear
  - 

- Use the previous method to show the top 3 words in King Lear, case-insensitive, and the number of times they appear.
  - 

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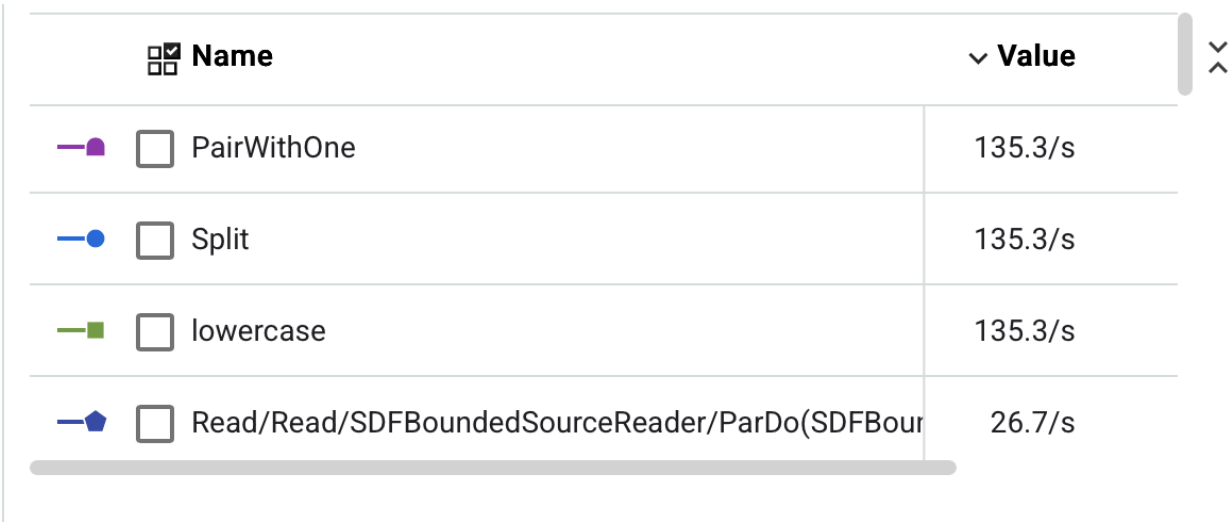
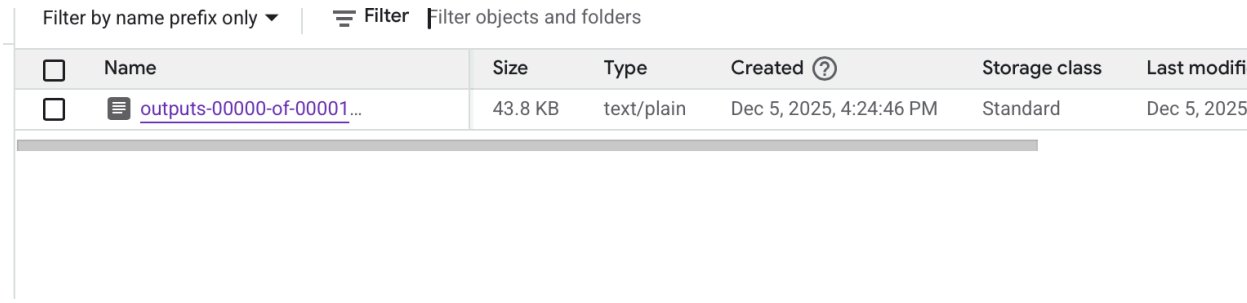
- The part of the job graph that has taken the longest time to complete.



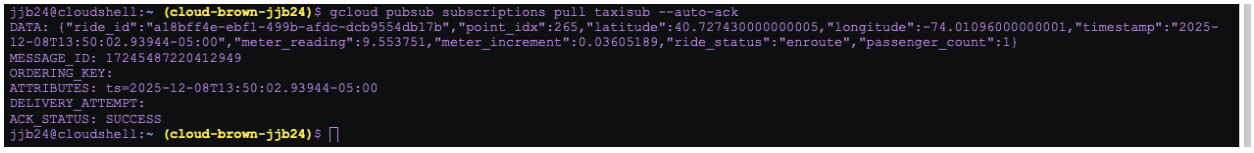
Name	Value
PairWithOne	135.3/s
Split	135.3/s
lowercase	135.3/s
Read/Read/SDFBoundedSourceReader/ParDo(SDFBour	26.7/s

- The autoscaling graph showing when the worker was created and stopped.

- Examine the output directory in Cloud Storage. How many files has the final write stage in the pipeline created?
  - 1

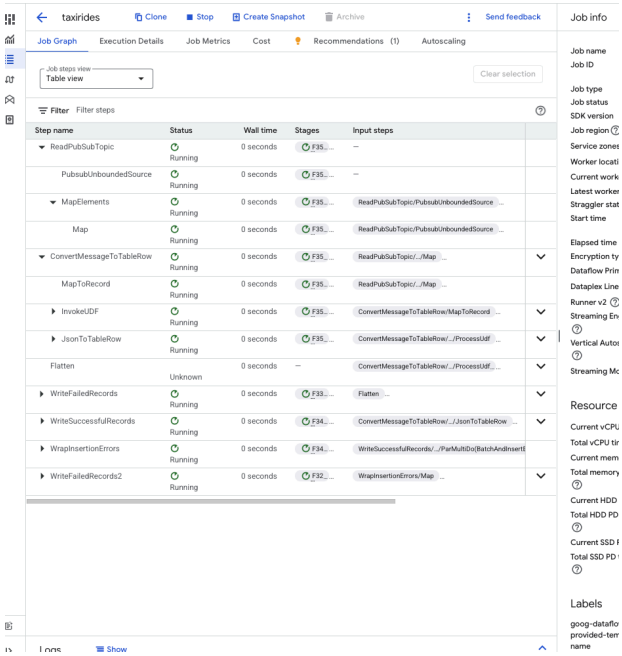


- Take a screenshot listing the different fields of this object.

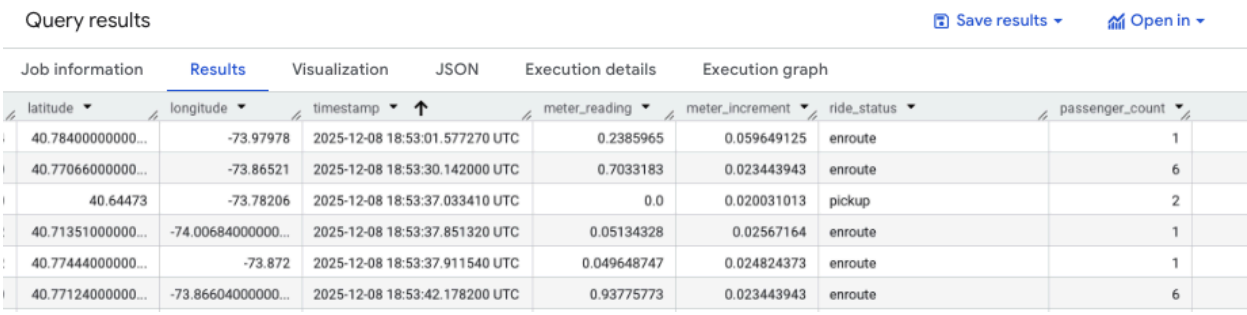


```
jjb24@cloudshell:~ (cloud-brown-jjb24)$ gcloud pubsub subscriptions pull taxisub --auto-ack
DATA: {"ride_id":"a18bff4e-ebf1-499b-afdc-dcb9554db17b","point_idx":265,"latitude":40.727430000000005,"longitude":-74.01096000000001,"timestamp":"2025-
12-08T13:50:02.93944-05:00","meter_reading":9.553751,"meter_increment":0.03605189,"ride_status":"enroute","passenger_count":1}
MESSAGE_ID: 17245487220412949
ORDERING_KEY:
ATTRIBUTES: ts=2025-12-08T13:50:02.93944-05:00
DELIVERY_ATTEMPT:
ACK_STATUS: SUCCESS
jjb24@cloudshell:~ (cloud-brown-jjb24)$
```

- Take a screenshot of the pipeline that includes its stages and the number of elements per second being handled by individual stages.



- Take a screenshot showing the number of passengers and the amount paid for the first ride



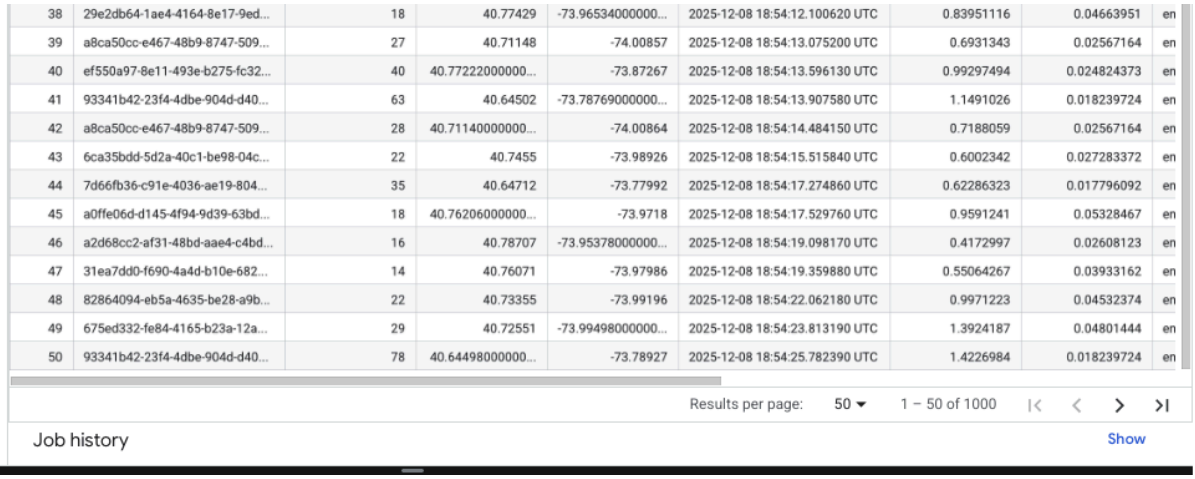
Query results

Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

latitude	longitude	timestamp	meter_reading	meter_increment	ride_status	passenger_count
40.78400000000...	-73.97978	2025-12-08 18:53:01.577270 UTC	0.2385965	0.059649125	enroute	1
40.77066000000...	-73.86521	2025-12-08 18:53:30.142000 UTC	0.7033183	0.023443943	enroute	6
40.64473	-73.78206	2025-12-08 18:53:37.033410 UTC	0.0	0.020031013	pickup	2
40.71351000000...	-74.00684000000...	2025-12-08 18:53:37.851320 UTC	0.05134328	0.02567164	enroute	1
40.77444000000...	-73.872	2025-12-08 18:53:37.911540 UTC	0.049648747	0.024824373	enroute	1
40.77124000000...	-73.86604000000...	2025-12-08 18:53:42.178200 UTC	0.93775773	0.023443943	enroute	6

- Take a screenshot showing the estimated number of rows in the table.



38	29e2db64-1ae4-4164-8e17-9ed...	18	40.77429	-73.96534000000...	2025-12-08 18:54:12.100620 UTC	0.83951116	0.04663951	en
39	a8ca50cc-e467-48b9-8747-509...	27	40.71148	-74.00857	2025-12-08 18:54:13.075200 UTC	0.6931343	0.02567164	en
40	ef550a97-8e11-493e-b275-fc32...	40	40.77222000000...	-73.87267	2025-12-08 18:54:13.596130 UTC	0.99297494	0.024824373	en
41	93341b42-23f4-4dbe-904d-d40...	63	40.64502	-73.78769000000...	2025-12-08 18:54:13.907580 UTC	1.1491026	0.018239724	en
42	a8ca50cc-e467-48b9-8747-509...	28	40.71140000000...	-74.00864	2025-12-08 18:54:14.484150 UTC	0.7188059	0.02567164	en
43	6ca35bdd-5d2a-40c1-be98-04c...	22	40.7455	-73.98926	2025-12-08 18:54:15.515840 UTC	0.6002342	0.027283372	en
44	7d66fb36-c91e-4036-ae19-804...	35	40.64712	-73.77992	2025-12-08 18:54:17.274860 UTC	0.62286323	0.017796092	en
45	a0ffe06d-d145-4f94-9d39-63bd...	18	40.76206000000...	-73.9718	2025-12-08 18:54:17.529760 UTC	0.9591241	0.05328467	en
46	a2d68cc2-af31-48bd-aae4-c4bd...	16	40.78707	-73.95378000000...	2025-12-08 18:54:19.098170 UTC	0.4172997	0.02608123	en
47	31ea7dd0-f690-4a4d-b10e-682...	14	40.76071	-73.97986	2025-12-08 18:54:19.359880 UTC	0.55064267	0.03933162	en
48	82864094-eb5a-4635-be28-a9b...	22	40.73355	-73.99196	2025-12-08 18:54:22.062180 UTC	0.9971223	0.04532374	en
49	675ed332-fe84-4165-b23a-12a...	29	40.72551	-73.99498000000...	2025-12-08 18:54:23.813190 UTC	1.3924187	0.04801444	en
50	93341b42-23f4-4dbe-904d-d40...	78	40.64498000000...	-73.78927	2025-12-08 18:54:25.782390 UTC	1.4226984	0.018239724	en

Results per page: 50 | 1 – 50 of 1000

Job history | Show

- Take a screenshot showing the per-minute number of rides, passengers, and revenue for the data collected

Query completed

Query results

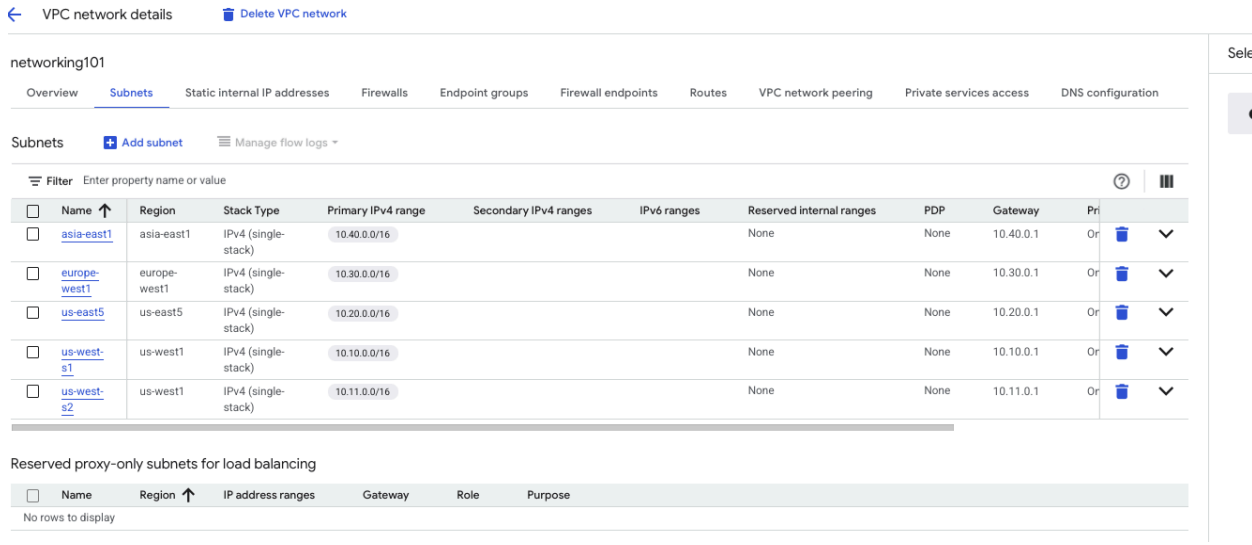
Job information | Results | Visualization | JSON | Execution details | Execution graph

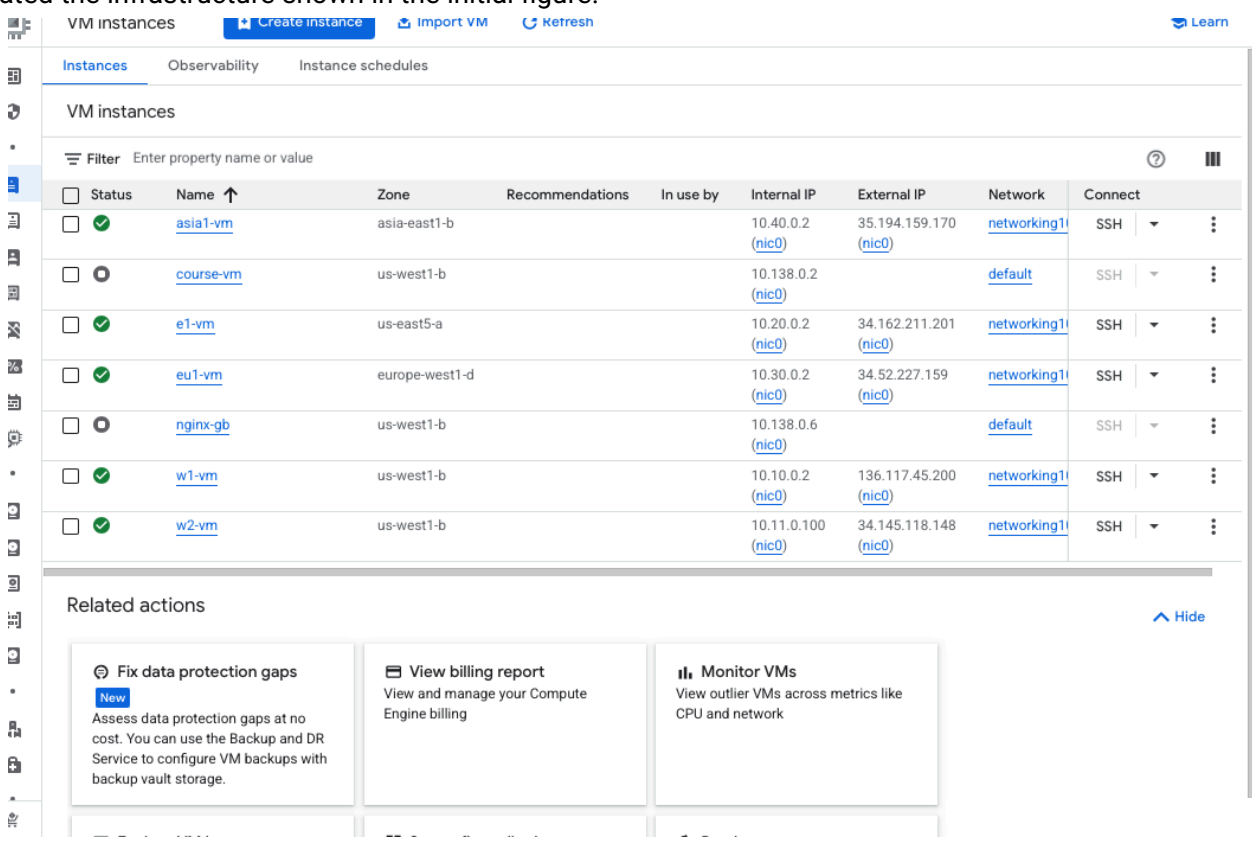
Row	minute	total_rides	total_passengers	total_revenue
1	10:49	2	2	4.3
2	10:50	6	8	51.15
3	10:51	352	568	4922.809997400...
4	10:52	510	887	7339.469992599...
5	10:53	500	809	7225.8999924
6	10:54	453	725	6713.390010400...
7	10:55	296	465	4342.810001600...

10.3g

- Take a screenshot of the output to include in your lab notebook. How many networks, subnetworks, and VM instances have been created?
  -

```
google_compute_instance.asia1_vm: Creation complete after 27s [id=pr

Apply complete! Resources: 11 added, 0 changed, 0 destroyed.

Outputs:

instance_internal_ips = {
  "asia1" = "10.40.0.2"
  "e1" = "10.20.0.2"
  "eu1" = "10.30.0.2"
  "w1" = "10.10.0.2"
  "w2" = "10.11.0.100"
}
instance_nat_ips = {
  "asia1" = "35.194.159.170"
  "e1" = "34.162.211.201"
  "eu1" = "34.52.227.159"
  "w1" = "136.117.45.200"
  "w2" = "34.145.118.148"
}
jjb24@cloudshell:~/cs430-src/10_CDN (cloud-brown-jjb24)$
```

- Visit the web console for VPC network and show the network and the subnetworks that have been created. Validate that it has created the infrastructure in the initial figure. Note the lack of firewall rules that have been created.
  - 

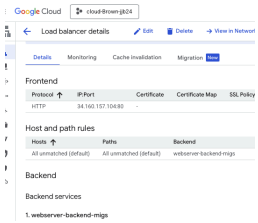
- Visit the web console for Compute Engine and show all VMs that have been created, their internal IP addresses and the subnetworks they have been instantiated on. Validate that it has created the infrastructure shown in the initial figure.
  - 

- Click on the ssh button for one of the VMs and attempt to connect. Did it succeed?
  - No it failed due to the firewall issues
- Take a screenshot that indicates the new rules have been deployed

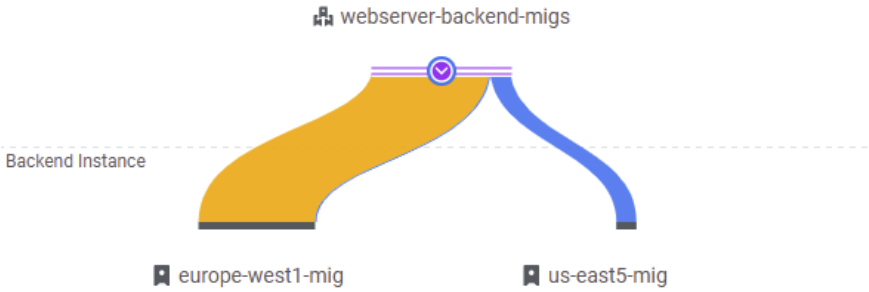
- Take a screenshot that indicates the new rules have been deployed

```
Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

Outputs:

instance_internal_ips = {
  "asia1" = "10.40.0.2"
  "e1" = "10.20.0.2"
  "eu1" = "10.30.0.2"
  "w1" = "10.10.0.2"
  "w2" = "10.11.0.100"
}
instance_nat_ips = {
  "asia1" = "35.194.159.170"
  "e1" = "34.162.211.201"
  "eu1" = "34.52.227.159"
  "w1" = "136.117.45.200"
  "w2" = "34.145.118.148"
}
jjb24@cloudshell:~/cs430-src/10_CDN (cloud-brown-jjb24)$
```

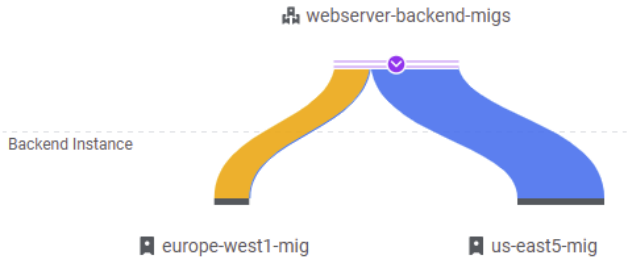
Location pair	Ideal latency	Measured latency
us-west1 us-east5	~45 ms	~60 ms
us-west1 europe-west1	~93 ms	~115 ms
us-west1 asia-east1	~114 ms	~140 ms
us-east5 europe-west1	~76 ms	~90 ms
us-east5 asia-east1	~141 ms	~170 ms
europe-west1 asia-east1	~110 ms	~135 ms

- Are the instances in the same availability zone or in different ones?
    - They are in different availability zones.
- List all availability zones that your servers show up in for your lab notebook.
    - us-east5
    - europe-west1
- Show a screenshot of the page that is returned.
    - 

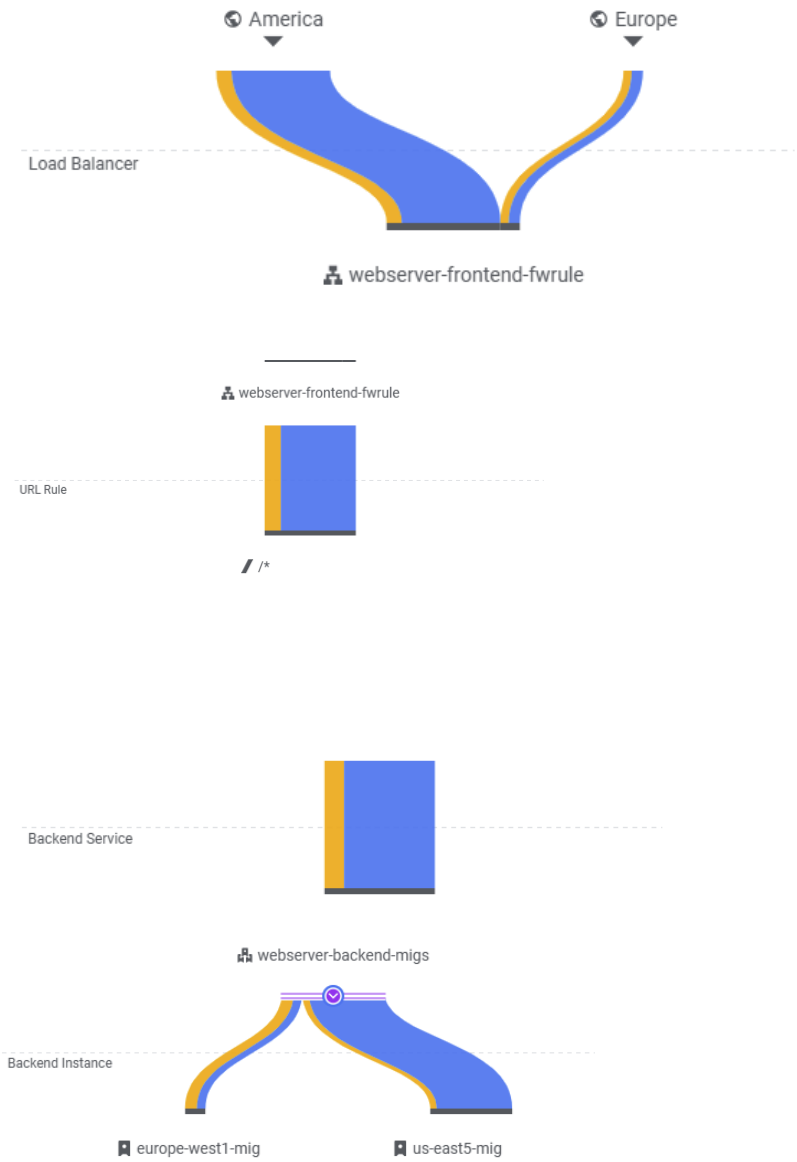
- Which availability zone does the server handling your request reside in?
    - us-east
- Does the backend server handling the request change?
    - Yes
- Take a screenshot of the initial traffic distribution



- Take a screenshot of the UI as additional instances are brought up and show that the traffic distribution shifts



- Show a screenshot of the final traffic distribution.


America
Europe
Load Balancer
webserver-frontend-fwrule
webserver-frontend-fwrule
URL Rule
/*
Backend Service
webserver-backend-migs
Backend Instance
europe-west1-mig
us-east5-mig